COLLEGE OF
ENGINEERING AND
COMPUTER SCIENCE

ECE 520/L

System-on-Chip Design with Lab

Final Project Report
Real–Time Signal Detection

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Introduction

This project implements a real-time sine wave generation and capture system on the Zybo Z7-10 development board using the Zynq SoC architecture. Four sine waves of varying frequencies and amplitudes are stored in block RAM (BRAM) on the programmable logic (PL) and selectively output based on user switch input. The captured samples are transferred to the ARM processing system (PS) via AXI GPIO interfaces, streamed over UART to a PC, and plotted using a Python script. The design demonstrates hardware-software co-design by handling signal generation entirely in the FPGA fabric while delegating communication and data transfer to the ARM processor.

Components Used

Description	Quantity	Manufacturer
Zybo Z7 ARM/FPGA SoC Board	1	Digilent
USB-A to Micro-USB Cable	1	N/A

Analysis and Results

The system successfully captured and transmitted 1,280 samples for each of the four sine waves when selected. The Vitis serial terminal confirmed correct UART transmission, displaying signed integer values within the expected amplitude ranges for each switch. The Python script successfully received the samples through an available COM port, saved them to a CSV file, and generated a waveform plot. The multi-switch error detection also functioned correctly, printing the appropriate error message whenever more than one switch was activated simultaneously.

Several challenges were encountered during implementation. The AXI GPIO blocks were automatically locked to a 4-bit width due to the Zybo Z7 board preset, requiring manual intervention to set the correct widths. A critical bug in the *sample_controller* module caused the divider counter to overflow since it was declared as a 10-bit register but needed to count to 50,000. Additionally, the PL logic was not receiving a proper reset signal from the Zynq PS, which required adding *rst_n* to *top.v* and connecting it to the processor system reset block in the block design.

The following figures show our results of the Real–Time Signal Detection system. Figure 1 shows the waveform corresponding to SW0 being switched on. Figure 2 shows the waveform corresponding to SW1 being switched on. Figure 3 shows the waveform corresponding to SW2 being switched on. Finally, Figure 4 shows the corresponding waveform to SW3 being switched on.

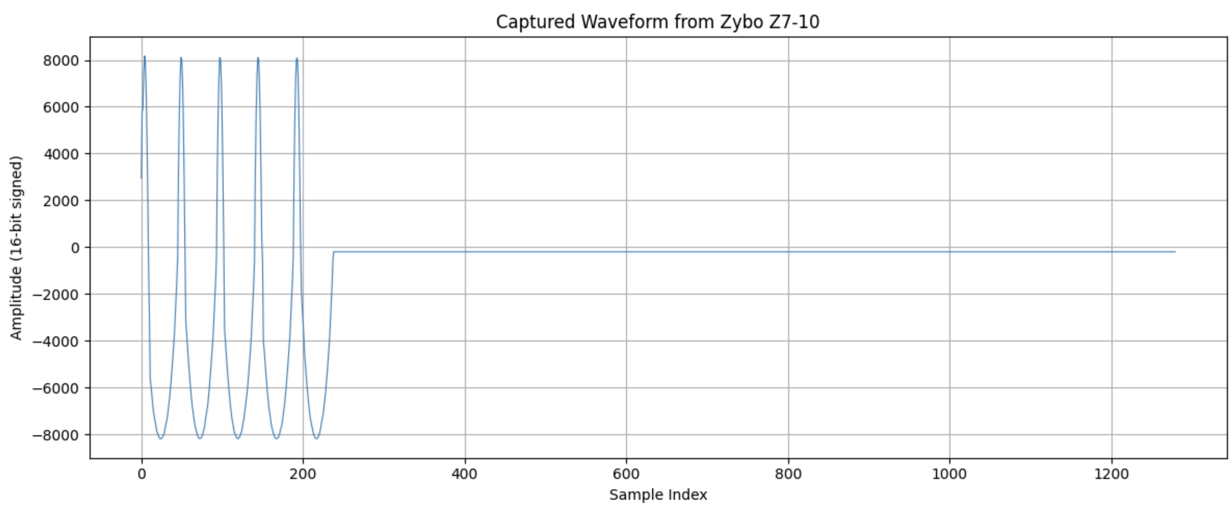

Fig. 1 - SW0 Signal Generated

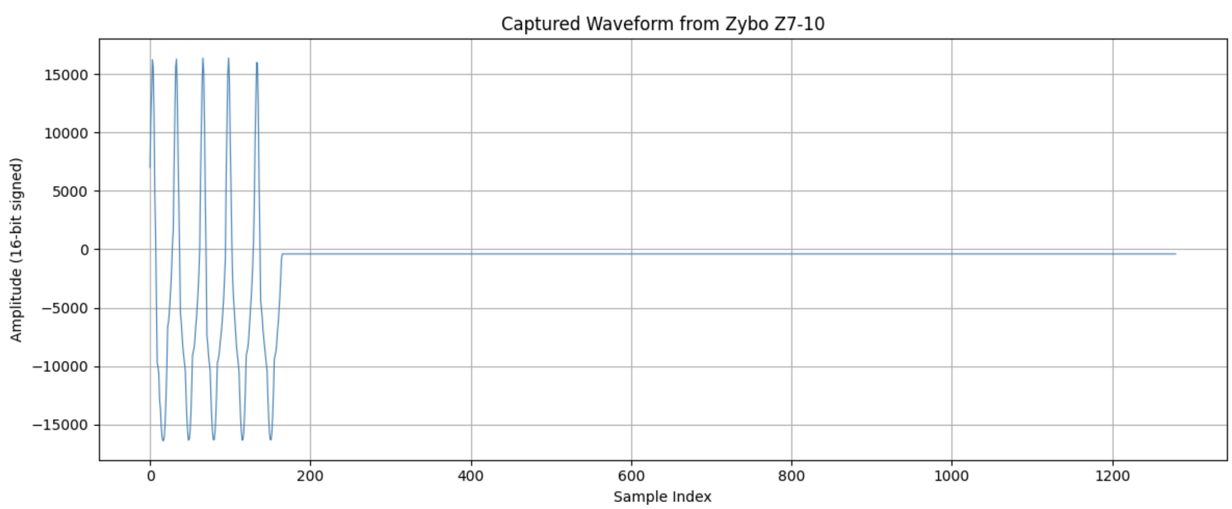

Fig 2. SW1 Signal Generated

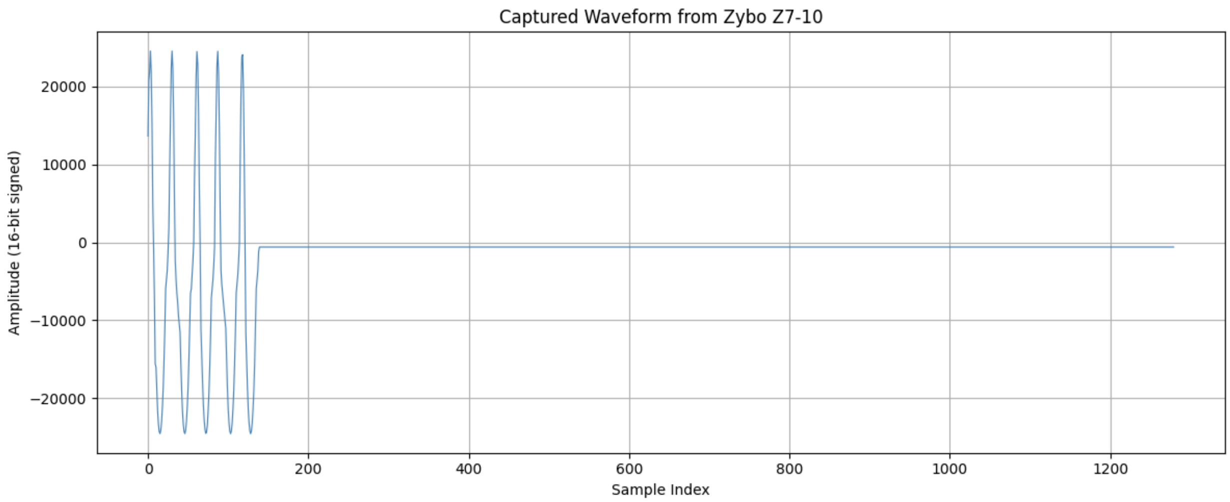

Fig 3. SW2 Signal Generated

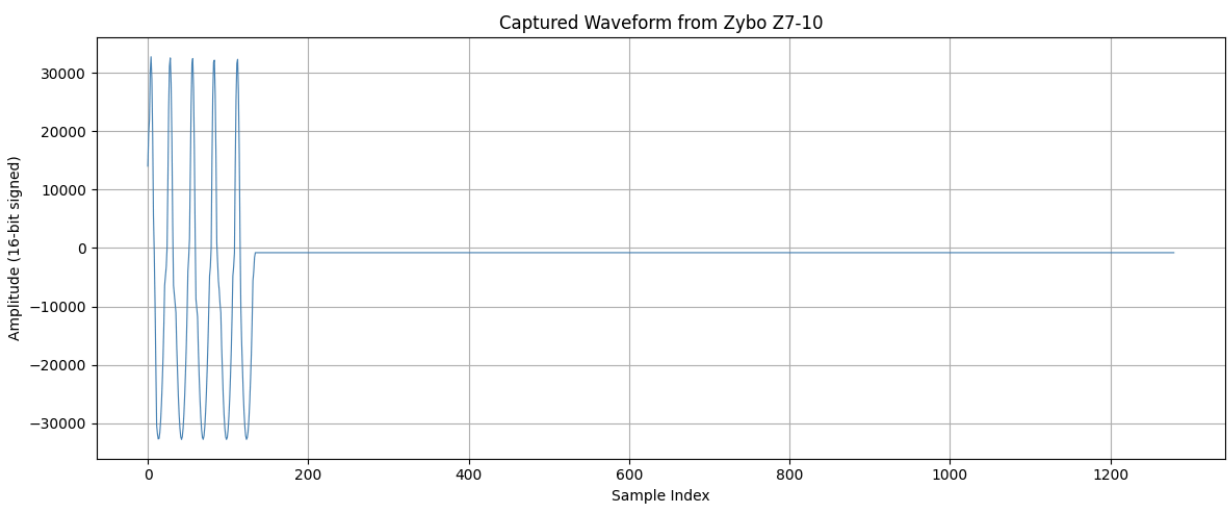

Fig 4. SW3 Signal Generated

Conclusion

This project reinforced the concepts of hardware-software co-design on the Zynq SoC platform, demonstrating how the programmable logic and processing system can be divided by responsibility to efficiently accomplish a shared task. Working through the design provided hands-on experience with AXI GPIO interfaces as a communication bridge between the PL and PS, and highlighted the importance of proper reset signal management when integrating custom RTL modules into a block design. The project also demonstrated how FPGA-based signal systems can interface with standard PC software tools for real-world data capture and visualization.

References

- **Demo Video Link**
- **GitHub Repository Link**